Computer Algorithms I

CS 401/MCS 401 14209, 14470, 15053, 15054

3 undergraduate hours, 4 graduate hours

Summer 2026, MWF 10am-11:40am, online via zoom

About Your Instructors

Instructor Name: Jan Verschelde (You may call me Professor Jan.)

Office: SEO 1210

Office Phone: N/A

Drop-in (Office) Hours: From 4pm to 5pm on Mondays, Wednesdays, and Fridays; or by appointment. Virtual via https://uic.zoom.us/my/profjanofficehour.

Email: janv@uic.edu.

TA Grader: Howard Chen (Email: hchen221@uic.edu). Office hours: Tuesdays from 11am to 1pm, virtually via https://uic.zoom.us/j/83708190049?pwd=0dSbj3beIwTjqabTga3iZ67Ati1i5a.1

Methods of Instruction: The lectures will be delivered via Zoom.

Course Description

Design and analysis of computer algorithms. Divide-and-conquer, dynamic programming, greedy method, backtracking. Algorithms for sorting, searching, graph computations, pattern matching, NP-complete problems.

What You Will Learn in This Course

Course Goals and Student Learning Outcomes

The goals of the course are to solve discrete problems in computer science. The learning objectives are to study the design and analysis of algorithms; to reason mathematically about the correctness and the execution costs.

Pre- or Co-requisites For This Class

Knowledge of data structures is the main prerequisite for this course.

Grade of C or better in MCS 360; or Grade of C or better in CS 251.

Course Website

The course web site is at https://homepages.math.uic.edu/~jan/mcs401 and a backup site is at https://janverschelde.github.io/mcs401/

On the course web site, the syllabus, lecture slides and homework assignments are posted. In this course, there will be some minimalistic use of canvas. Gradescope will be used to collect all answers to homework and exams.

https://piazza.com/uic/summer2026/mcs401 is the piazza web site where you can post questions and read answers to the questions.

Course Texts and Materials

The textbook for this course is *Algorithm Design* by Jon Kleinberg and Eva Tardos, Pearson, 2006.
Slides used in the lecture are based on the textbook, which provides an extended narrative.

Required Technology

As the course runs online, ensure you have good internet access.

Required Reading, Video, or Lab Materials

Summer of 2024, pre-recordings of the lectures were posted at youtube, available as a playlist at https://www.youtube.com/playlist?list=PLGMCOm5grOW7BZ2jQt7TAmQ5eAEiOsP8p

These videos are still relevant, although the current lectures may have some substantial modifications.

Recommended (Optional) Texts or Other Materials

For some topics, we will refer to the textbook *Introduction to Algorithms* (Third Edition, MIT Press, 2009) of Thomas H. Cormen, Charles E. Leierson, Ronald L. Rivest, and Clifford Stein.

Our Inclusive Learning Environment

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If there are aspects of the instruction or design of this course that result in barriers to your inclusion, engagement, accurate assessment or achievement, please notify me as soon as possible.

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, you should connect with the ***Disability Resource Center (DRC) at drc.uic.edu, drc@uic.edu, or at (312) 413-2183*** *to create a plan for reasonable accommodations. In order to receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Letters of Accommodations can be presented at any point of the semester and,* *unless otherwise noted, do not expire once obtained. Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.*

Community Agreement

Community Agreement: Ground Rules for a Safe/Brave Space

- *Be present (turn off cell phones and remove yourself from other distractions)*
- *Be respectful*
- *Assume goodwill*
- *Try not to make assumptions; seek to understand, not to judge*
- *Be open to challenges as an opportunity to learn something new*
- *Be flexible when things don't work*
- *Share helpful tips*
- *No side conversations*
- *Be willing to work together*
- *Be mindful of one another's privacy -- do not invite outsiders into our classroom*

In the zoom lectures, feel free to type in questions and comments in the chat window.

How Your Learning Will Be Assessed

There will be six homework collections, one midterm exam, and a final exam.

Selected homework problems are assigned with each lecture but is recommended that you try all homework problems. Homework will be collected every week, except during exam weeks. The main goal of the lectures is to get you started with the solution of those problems. As homework counts for only 100 out of the total 400 points, and is thus a low-stakes assessment, it counts as important practice for what is expected as correct answers on the high-stakes exams.

During the semester, there will be one midterm exam worth 100 points. No makeup for the midterm exam will be given. If the midterm exam is missed, then greater weight will be placed on the final exam, especially on the material covered on the missing exam.

The final exam counts for 200 out of the 400 points and is therefore mandatory for passing the course. As the final exam happens on the last day of this 8-week summer course, there will be very few opportunities for rescheduling.

Assessments

Your final grade will be computed as the percentage of the 400 points: 100 points from homework, 100 from the midterm exam, and 200 points from the final exam.

Grading scale

A	90% - 100%
B	80% - 89%
C	70% - 79%

D	60% - 69%
F	0% - 59%

Student Accountability for Coursework

In this online course, you should have good internet access and be aware of the Chicago time if you are living in another time zone.

During exams, a ten-minute grace period may be typically included to allow for internet delays. Please do not use this grade period to continue working on the exam questions. The online system allows for multiple submissions. Submit your answers as soon as you have finished.

When requesting an extension of a deadline to submit homework, you may expect to receive a couple of hours, not days or a week. Submitting a partial answer to homework is better than submitting nothing.

Midterm Grades

There is no formal midterm grade. By the day of the midterm exam, after four weeks of classes, you should have a good understanding of first half of the materials and to a good grade in the course.

If after four weeks of classes you still feel lost in the course, then please reach out to me for help.

Final Exam

The final exam is cumulative, covering all topics in the course.

Course Policies

Tips for Success

When learning a new algorithm as a computer scientist ask yourself the following question: can I write detailed instructions for a computer to execute this new algorithm? While you will not be asked to produce working code in this course, asking this main question should guide your learning of any new algorithm.

There are many ways for you to show us what you know or can do and how you are learning, through your effort, interactions, and class participation, application of critical thinking skills to solve real world problems, and performance on assignments and exams. This class has been structured to help all students get the support and guidance needed to succeed in your learning. The following tips will help you be successful in this class and in other classes throughout your career at UIC.

- *Ask questions during class.*
- *Complete assignments on time.*
- *Participate in class discussions.*
- *Individually review your notes after class.*
- *Post questions on the discussion board or ask questions during drop-in (office) hours about concepts or procedures that seemed confusing.*
- *Attend / engage in drop-in (office) hours with me and your TA.*

Below are some additional tips specific to online learning.

- *Have practically unlimited access to a computer and internet services*
- *Commit to the specified amount of time {average hours/week} to the course*
- *Be self-motivated and self-disciplined, keeping up with assignments and completing coursework on time*
- *Communicate through writing as well as use video and audio methods as needed for oral communication*
- *Think through ideas before responding*
- *Accept critical thinking and decision-making as part of the learning process*
- *Speak up if problems arise because non-verbal communication is not possible in an online setting*
- *Embrace a mindset in which you believe high-quality learning can take place without going to a traditional classroom.*

Finally, do the exercises! They are the low-stakes assignments in this course. Many of the exercises help you to become familiar with the algorithms and prepare you for the high-stakes exams.
Read the textbook. The main goal of the lectures is to provide you with better access to the literature.

Attendance

Students are expected to attend all class meetings. Any changes in this syllabus or in the scheduling of exams and other assignments will be announced during class meetings.

An online course offers more flexibility (no commute) but requires more self-discipline. Prepare to attend as the class was in person and secure yourself in a quiet place without distractions.

The summer course is a compressed version of a regular semester course, compressed in the sense that each of our sessions counts for two lectures in a regular course. Missing two consecutive sessions is the equivalent of a long absence and is likely leading to missing out on essential materials.

Religious Holidays

I will make every effort to avoid scheduling exams or requiring student projects be submitted on religious holidays. If you wish to observe your religious holidays, please notify me by the tenth day of the semester of the date when you will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, please notify me at least five days in advance of the date when you will be absent. I will make every reasonable effort to honor your request and not penalize you for missing the class. If an examination or project is due during your absence, you will be given an exam or assignment equivalent to the one completed by those students in attendance. Students may appeal through campus grievance procedures for religious accommodations.

Class Participation / Engagement

Because active participation in the course is one of the most important ways to learn, you will be assessed on your class engagement. Engagement looks a little different for everyone, but in general, an engaged student will come to class prepared, contribute regularly to class activities or discussions, listen attentively to peers and the instructor, stay on-task during class, complete their work in a timely manner, and reach out to the instructor if they have questions or start to fall behind. If you anticipate any barriers to your full engagement in the course, I encourage you to contact me so we can strategize about how you can best fulfill the course requirements.

Your engagement or lack thereof will not be assessed.

Privacy Notification and Policy for Video Recording of Class Sessions

The recordings during the online sessions focus on the progression and narratives of the slides.

While chat discussions may contain identifying information, note that they will be available only to those enrolled in the course.

Classroom Conduct Policy

- *Be in a quiet place and use headphones*
- *Mute the microphone unless talking*
- *Video on when speaking or engaging in group activities*
- *Quit all other screens (email, text, social media) to create a classroom "presence" (also helps with internet stability)*
- *Actively participate in class*
- *Raise your hand if you have a question to ask verbally*
- *Use the chat feature if you prefer to type your question*

Policy for Late or Missing Work

Deadlines may be postponed. If you know you will be late, then it is better to apply for an extension of the deadline, instead of not submitting anything. For missed assignments, greater weight may be placed on the final exam.

Policy for Revisions

You will not be allowed to resubmit revised homework for a better grade.

Policy for Regrading

Any request for a regrade must be made within one week of the assignment being returned to you. If you think there has been a simple arithmetic error on your assignment then write a note explaining the error, attach this to the front of the assignment, and turn it into your TA. If the error is confirmed, then the points will be added to your score before the end of the semester. If you think there was an error in grading that is not an arithmetic error, write a brief note explaining why you think more points should be awarded, include it with the assignment, and submit it to your TA. If the points you request will affect your final grade, then I will reevaluate the assignment for the contested points.

Academic Integrity

No student shall claim or submit the work of another as one's own.
Allowing someone to copy from you is also punishable. All your work in this class will be individual.
Verbatim copying the output of generative AI tools is plagiarism.

UIC is an academic community committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. In this community, all members including faculty, administrators, staff, and students alike share the responsibility to uphold the highest standards of academic honesty and quality of academic work so that such a collegial and productive environment exists.

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all of your academic endeavors. When accusations of academic dishonesty occur, the Office of the Dean of Students investigates and adjudicates suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct. Please review the UIC Student Disciplinary Policy for additional information about the process by which instances of academic misconduct are handled towards the goal of developing responsible student behavior.

By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not communicated with anyone in any way during an exam or other online assessment. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

Use of Generative Artificial Intelligence

In response to my query "use of generative ai to computer algorithm" the AI overview of Google started its reply with "Using generative AI is a highly effective way to learn computer algorithms because it acts as an interactive, 24/7 personalized tutor." giving among its references an insightful 2024 paper in the Communications of the ACM on "Computing Education in the Era of Generative AI."

Using generative AI to help you learn is good as long as you learn, that is: you gain understanding and insight in the design and analysis of algorithms. Be aware of the disclaimer: "AI can make mistakes, so double-check responses."

FERPA

Federal law (FERPA) prohibits me from disclosing or discussing your grades over e-mail. Please come see me during drop-in hours or make an appointment if you want to talk about your work.

Course Schedule

Below is the outline of the course. The section numbers refer to the textbook.

Week 1 – *introduction and the Gale-Shapley algorithm*

- *L-1 Mon 15 June:* The Gale-Shapley algorithm to solve the stable matching problem; five representative problems (sections 1.1 and 1.2)
- *L-2 Wed 17 June:* cost and complexity -- Gale-Shapley revisited, measuring algorithm complexity, efficient algorithms, asymptotics, big-O, big-Omega, and big-Theta (2.1, 2.2)
- *Fri 19 June:* Juneteenth holiday. No classes.

Week 2 – *graph algorithms*

- *L-3 Mon 22 June*: graph algorithms -- implementing and analyzing the Gale-Shapley algorithm, running times, graph representations and traversals (2.3, 2.4, 3.1, 3.2)
- *L-4 Wed 24 June*: graph traversals -- implementing breadth first search and depth first search with queues and stacks, connectivity and bipartiteness testing (3.3, 2.5, 3.4)
- *L-5 Fri 26 June:* greedy algorithms -- directed graphs, representations and search algorithms,

topological ordering of directed acyclic graphs, greedy algorithms for interval scheduling, exchange arguments, the interval partitioning problem (3.5, 3.6, 4.1, 4.2)

Week 3 – *end of graph algorithms – start of divide and conquer*

- *L-6 Mon 29 June:* scheduling by minimizing maximum lateness, optimal caching: minimizing cache misses, shortest paths in weighted directed graphs by Dijkstra's algorithm (4.3, 4.4)
- *L-7 Wed 1 July:* spanning trees -- three greedy algorithms: Kruskal's and Prim's algorithms, reverse-delete algorithm; Union-Find and priority queues (4.4, 4.5, 4.6)
- *Fri 3 July:* Independence Day holiday. No classes.

Week 4 – *divide and conquer -- review for final exam*

- *Mon 6 July:* introduction to divide and conquer via Merge sort, counting inversions, finding the closest pair of points in the plane, Karatsuba's integer multiplication (5.1, 5.2, 5.3, 5.4)
- *Wed 8 July:* FFT and Strassen -- Fast convolutions, the master method to solve recurrences, Strassen's method to multiply matrices (5.6)
- *Fri 10 July:* review for the midterm exam

Week 5 – *midterm exam – dynamic programming*

- *Mon 13 July:* midterm exam
- *Wed 15 July:* introduction to dynamic programming, memoization, and bottom-up recursion, weighted interval scheduling (6.1, 6.2)
- *Fri 17 July:* introduction to least squares, segmented least squares, dynamic programming with 2D tables, the sequence alignment problem (6.2, 6.3, 6.6)

Week 6 – *network algorithms*

- *Mon 20 July*: shortest paths and knapsack -- Bellman-Ford dynamic programming algorithm, subset sum & knapsack problems, max flow, Ford-Fulkerson algorithm (6.4, 6.8, 7.1)
- *Wed 22 July:* max-flows and min-cuts in a network, choosing good augmenting paths (7.2, 7.3)
- *Fri 24 July*: bipartite matching problem, extensions of max flow (7.5, 7.7)

Week 7 – *more scheduling -- introduction to NP versus P*

- *Mon 27 July:* survey design, airline scheduling problems (7.8, 7.9)
- *Wed 29 July:* introduction to computational complexity and polynomial-time reductions, vertex cover and independent set, gadget reductions from 3SAT (8.1, 8.2)
- *Fri 31 July:* definition of NP; SAT, CIRCUIT-SAT, and NP completeness (8.3, 8.4)

Week 8 – *Hamiltonian Path and TSP -- review & final exam*

- *Mon 3 August:* sequencing problems: Hamiltonian path and traveling salesperson (8.5) additional NP-Complete problems, co-NP and the asymmetry of NP (8.6, 8.8, 8.9)
- *Wed 5 August:* review for the final exam
- *Fri 7 August:* final exam

Academic Deadlines

Important dates in the semester are listed below.

- Thursday 18 June: Homework one is due at 10am.
- Friday 19 June: Juneteenth holiday. No classes.
- Friday 26 June: Homework two is due at 10am.
- Thursday 2 July: Homework three is due at 10am.
- Friday 3 July: Independence Day holiday. No classes.
- Friday 10 July: Last day for an optional late drop.
- Friday 10 July: Homework four is due at 10am.

- Monday 13 July: Midterm exam.
- Wednesday 22 July: Homework five is due at 10am.
- Friday 31 July: Homework six is due at 10am.
- Friday 7 August: Final exam.

Providing Feedback About This Course

At the end of the term, you will be asked to complete the course evaluations.

UIC Resources - Supporting Student Health, Wellbeing, and Academic Success

As a student at UIC, you may experience challenges such as struggles with academics, finances, student life, or your personal well-being. Please know this is completely normal and that you shouldn't hesitate to ask for help. Come to me, or if it is about an issue beyond the scope of this class, please contact your college advisors, or get help from any number of other support services and resources available to all UIC students:

- *Current Student Resources – a one-stop shop for links to resources in the following categories: General, Academic, Student Support, Student Life, Technology, Health and Safety, and Getting Around Campus*
- *UIC Tutoring Resources*
- *Programs and initiatives supporting the UIC undergraduate experience and academic programs through the Student Success Initiative*
- *If you are experiencing personal hardships (including but not limited to: personal and family emergencies, academics, interpersonal conflicts, personal safety, physical and mental health concerns, food or housing insecurity) and need resources and assistance, please reach out to the Student Assistance division, supported by UIC's Office of the Dean of Students.*
- *Student Guide for Information Technology – a comprehensive resource for UIC students describing the most commonly used IT services and tools supporting your success*
- *Navigating Health Resources at UIC – explore the many resources that are on campus to support students' health and wellbeing and learn about how you can access them.*

Importantly, if you are in immediate distress, please know the UIC Counseling Center has crisis intervention services available 24 hours a day, 7 days a week. You can walk into the Counseling Center during business hours to request to talk to a crisis counselor, or you can call the Counseling Center main line at (312) 996-3490. If you are calling outside of business hours, select "2" during the automated greeting to be connected to a crisis counselor. You can find additional mental health resources and services on the Counseling Center Website and the Student's Guide to Accessing Behavioral Health Services at UIC.

For immediate crisis support outside of UIC, call or text "9-8-8" any time, 24 hours a day, 7 days a week. Dialing 988 will connect you to a trained crisis professional. 988 offers services in Spanish, American Sign Language, and over 240 languages through tele-interpretation services. The 988 website has an online chat feature, too.

*The **Writing Center** offers friendly and supportive tutors who can help you with reading and writing in any of your courses, not just English. Tutors are ready to help with other writing as well, such as job applications, personal statements, and resumes. The tutor and you will work together to decide how to improve your writing. If you have not started your assignment, that is OK. A tutor can help you brainstorm or make an outline. Tutors understand that you might be using the Writing Center for the first time. They are ready to guide you through your first session. You can choose to work with a tutor in person or in real time over the internet using video, audio, or chat and a whiteboard. For more information and to schedule an appointment, visit the Writing Center website.*

*The **Math and Science Learning Center**, located in the Science and Engineering South Building (SES) at 845 W. Taylor St. 3rd Floor, Room 247, is a meeting place for students in Math, Biological Sciences, Chemistry, Earth and Environmental Sciences, and Physics. At the MSLC, students can meet with graduate teaching assistants for tutoring in 100-level courses, attend informal group study sessions with other students, or meet up with friends to attend one of the workshops, seminars, or other activities sponsored by the MSLC during the semester. Visit the MSLC website, call 312-355-4900, or email at mslc@uic.edu.*

*The **UIC Library** is located both on east and west campus, provides access to resources, study rooms, and research support both online via chat and in person. At Daley Library on the east side of campus, stop by the reference desk in the IDEA Commons, or make an appointment for research help on either side of campus. Learn more about library services and resources, and to find research materials in specific subject areas view the Research Guides.*

*The **Academic Center for Excellence (ACE)** can help if you feel you need more individualized instruction in reading and/or writing, study skills, time management, etc. Please call (312) 413-0031 or visit the Academic Center for Excellence (ACE) website for more information.*

***UIC Counseling Center -** Counseling services are available for UIC students who pay the student health services fee. You may seek a free and confidential intake with a trained therapist from the Counseling Center, who will work with you to create a clinically appropriate treatment plan. Treatment options may include Brief Individual Therapy or Group Therapy at the Counseling Center, referral to Psychiatry services, or Care Coordination to be connected to a community provider who meets your clinical needs. The Counseling Center is located in the Student Services Building. You may contact them at (312) 996-3490 during business hours to learn more about services, set up an intake, or for crisis intervention. Crisis intervention services are available 24/7 by walking into the Counseling Center during business hours or by calling. If you are calling after business hours, please select "2" when the automated voice message begins. For more information about the Counseling Center or accessing the Counseling Center and other mental and emotional health services on campus, check out the Student's Guide to Accessing Behavioral Health Services at UIC. For more information about services in the community, check out our Community Provider Database.*

*The **Campus Advocacy Network** provides information and offers resources to all UIC students, faculty, and staff. Under the Title IX law, you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim services and advocacy, contact UIC's Campus Advocacy Network at 312-413-8206. To make a report to UIC's Title IX office, email TitleIX@uic.edu or call (312) 996-8670.*

Campus Security - *As a UIC student, you've chosen to live in one of the nation's largest cities. But, as at any university, crime is a reality. At UIC, we are strongly committed to our public safety programs, and we encourage students to be proactive in learning what programs and services are available in case of an emergency. You are* ***discouraged*** *from staying in university buildings alone, including lab rooms, after hours and are* ***encouraged*** *to use the [Walking Safety Escort] and/or [Night Ride] if you are uncomfortable traveling anywhere on campus. Also, you can download [the UIC SAFE app], a free personal security tool for students, faculty, and staff. It allows you easy contact with dispatchers and first responders in case of an emergency. Finally, by dialing 5-5555 from a campus phone, you can summon Police or Fire for any on-campus emergency. You may also program the complete number, (312) 355-5555, on speed dial on your phone.*

Emergency Response Systems and Guides *- The emergency response guides can be found at the [Office of Preparedness and Response]. Please review and acquaint yourself with the guide and recommendations for various emergency situations.*

Student Health Services – *If you get sick during the semester, please be aware that there are resources on campus that can support you. The Family Medicine Center is committed to providing high-quality, student centered care and services. It has attending physicians, residents, and nurse practitioners on staff to provide the full range of primary health care services. The Family Medicine Center also offers a limited set of health services at* ***no additional cost to students who pay the Student Health Service Fee (SHSF)****. Services include caring for an acute or urgent illness (e.g., cold, strep throat, asthma attack, flu) or an injury. More information can be found on the [Family Medicine Center's Student Health] webpage.*

Supporting Parenting Students *- Balancing school, childcare, and work can be challenging, and we want you to know you're not alone. If you are a parenting student, we encourage you to share your situation early so we can support you and help troubleshoot any issues. Occasionally, bringing a child to class to cover unexpected gaps in care is perfectly acceptable—please sit near the door so you can step out if needed. Non-parenting classmates, please leave seats near the door available for this purpose.*

UIC is committed to supporting student parents through the Little Sparks Program, funded by the Child Care Access Means Parents In School (CCAMPIS) grant. Little Sparks offers affordable, high-quality childcare, lactation spaces, study-play areas, family events, and a dedicated coordinator to help parenting students succeed. For more information, visit [Little Sparks Program] or contact Melinda Young at [mmcghe2@uic.edu].

Disclaimer

This syllabus is intended to give guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance as in-class announcements and in writing via Blackboard Announcements.

Last Revised: Wednesday 17 June 2026